CIVICPULSE

Monitoring and Forecasting Pakistani Civic Issues

News-derived signals for mobility, electricity, water, and municipal services

CIVIC SIGNALS, CLEARLY

Monitoring report activity and experimental forecasts across Pakistan

The problem CivicPulse addresses

THE PROBLEM

Civic-service problems appear across many news sources, but the information is scattered and difficult to compare over time. CivicPulse brings these signals into one traceable workflow so people can explore where report activity appears and how it changes.

1 **Who needs the signal?**
Residents, service teams, planners

2 **Why it matters**
Scattered reports hide patterns

3 **What CivicPulse provides**
A shared view of report activity

Sources and dataset snapshot

Dataset	News reports + hourly weather features
Sources	FreeNewsAPI, sampled GDELT, Open-Meteo
Current snapshot	61,254 Bronze rows → 13,837 Silver articles
Gold output	142 issue events / hourly count rows
Coverage	GDELT: ~2 years sampled; weather: Karachi only
Key fields	City, issue, hour, report count, weather; count is target

61K+
Bronze rows

4
issue groups

1 city
weather coverage

How the data becomes useful

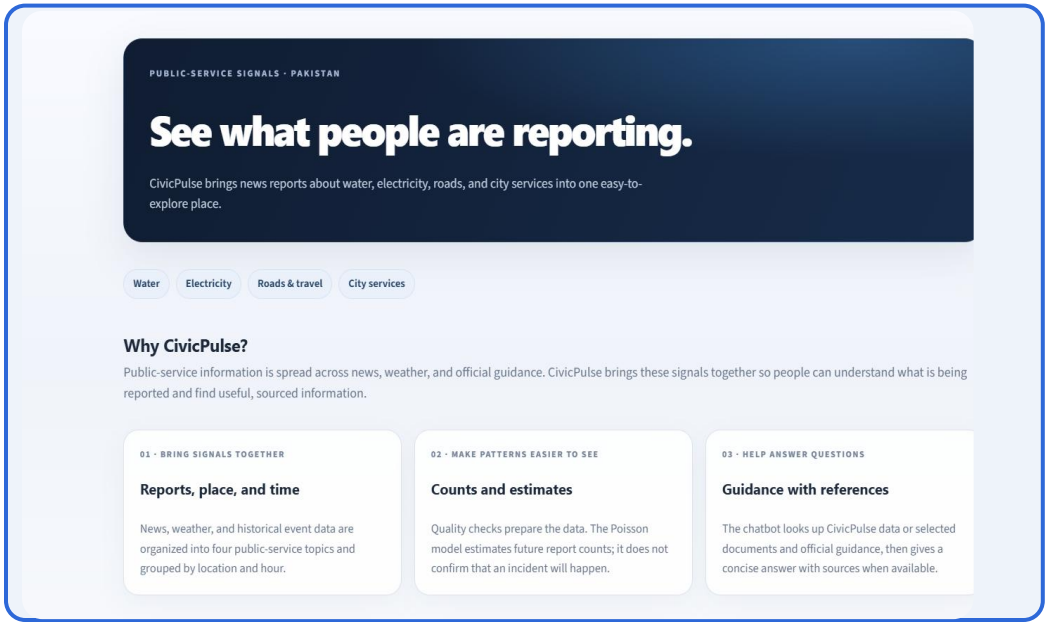


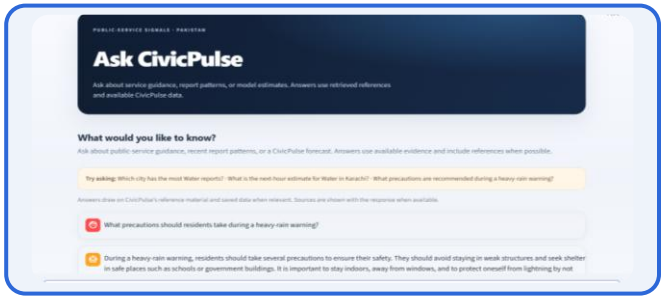
Raw sources → quality-checked counts → forecast evidence → clear, cited answers

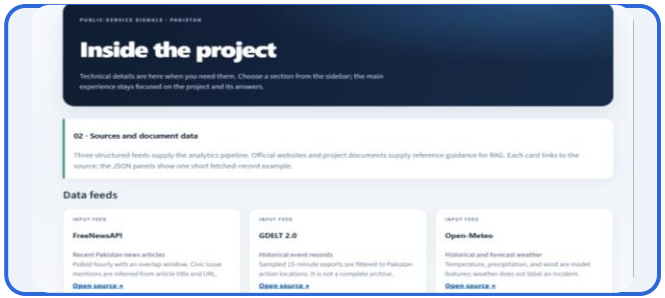
What runs at each stage

Stage	Tools	What it does
Collection	Python · APIs	Fetch recent news, sampled history, weather
Warehouse	DuckDB · pandas	Bronze → Silver → Gold; record quality actions
Forecast	Poisson · baseline	Estimate hourly report counts; compare errors
Retrieval	Chroma · embeddings	Find source passages and structured evidence
App + schedule	Streamlit · Supabase · Actions	Show data; sync Gold and forecasts hourly

CivicPulse interface







Streamlit app | Live values are read from Supabase

Results and current limits

1. **The pipeline is reproducible**

   A successful snapshot processed 61K+ raw rows.

2. **The forecast is experimental**

   Weekly baseline MAE was lower than Poisson MAE.

3. **Coverage needs improvement**

   Sparse Gold events; weather snapshot covers Karachi.

WHAT USERS GAIN

A traceable view of report patterns, with limits shown.

NEXT

Improve city normalization, coverage, and evaluation data.

CivicPulse team & credits

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Sources and model limits are documented in the project

Thank you